Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)
SEMESTER II EXAMINATION
BT705/MI705: IMMUNOLOGY

Date: 28.04.2010
Time: 10.00 AM to 1.00 PM
Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1. Complement is a chemical mediator that is involved in:
(a) lysis of microorganisms (b) facilitates phagocytosis
(c) innate immunity (d) all of the above

2. The following is **NOT** the characteristic of adaptive immunity
(a) antigenic specificity (b) diversity (c) pattern recognition (d)non self recognition

3. Complementarity determining regions (CDRs) are the Ag binding sites that are complementary to the structure of the:
(a) heptane (b) adjuvant (c) epitope (d) all of the above

4. The major difference between the monomeric and polymeric IgM is the presence of :
(a) J region (b) hinge region (c) variable region (d) all of the above

5. Virulence is independent of the following characteristic of a pathogen:
(a) invasiveness (b) infectivity (c) disease caused (d) pathogenic potential

6. The following is **NOT** the survival strategies of viruses into the specific host:
(a) long latent period (b) facile transmission (c) coexistence (d) lysis of host cell

7. After ingestion by alveolar macrophages, *M.tuberculosis* is able cause disease by:
(a) inhibiting phagolysosome formation (b)activation of macrophages
(c) production of cytokinesTH1 (d) all of the above

8. The classical complement pathway differs from alternative pathway as it requires the presence of the following:
(a) Ab-Ag complex (b) microorganism (c) lipopolysaccharides (d) all of the above

9. During a primary humoral response, the following immunoglobulin is secreted initially
(a) IgG (b) IgA (c) IgM (d) IgA

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10. Cells that display peptides associated with MHC class I molecules to CD8 T cells are referred to as:
(a) target cells (b) effector cells (c) antigen-presenting cells (d) none of the above

11. Presentation of nonpeptide (lipid and glycolipid) antigens derived from bacteria involves
(a) CD8 molecules (b) CD1 molecules (c) MHC class II (d) all of the above

12. Superantigen is a type of exotoxin that act by stimulating T cells directly to release
(a) cytotoxin (b) cytokines (c) glycoproteins (d) all of the above

13. The following is **NOT** an example of systemic auto-immune diseases:
(a) multiple sclerosis (b) scleroderma (c) rheumatoid arthritis (d) all of the above

14.The following autoimmune disease **DOES NOT** activate complement system:
(a) Rheumatoid arthritis (b) good pasture's syndrome
(c) diabetes mellitus (d) multiple sclerosis

15.Bronchial asthama is an example of :
(a) systemic anaphylaxis (b) atopic reaction (c) agglutination (d) all of the above

16. Allergic contact dermatitis is caused by the binding of the following molecules with proteins in the skin:
(a) adjuvants (b) antibodies (c) haptens (d) none of the above

17. The following type of vaccine is used for immunization against mumps:
(a) attenuated (b) killed (c) Endotoxin (d) none of the above

18. The following is **NOT** an important constituents of HAT media:
(a) Hypoxanthine (b) Thymidine (c)Aminopterin (d) none of the above

19. Cross-Reactivity occurs when different Ag have
(a) no epitopes are present (b) completely dissimilar epitopes present
(c) near identical epitopes (d) none of the above

20. Chickenpox is caused by:
(a) *Leptospira interogans* (b)Varicella virus (c) Rubella virus (d) Variola virus

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(PART B)

Section I

Q.1.A. Describe any two of the following: **(04)**

i. Activation of macrophages
ii. Types of dendritic cells
iii. Formation of heavy chains of Immunoglubulin by combinatorial joining

Q.1. B. Differentiate between any two of the following: **(04)**

i. Secreted and membrane bound antibodies
ii. Isotype and allotype
iii. Primary and secondary lymphoid follicles

Q.1.C. Explain why does swelling occurs in lymph nodes upon severe antigenic stimulation? **(02)**

Q. 2.A. Write short note on any one of the following: **(04)**

i. Mode of neutralization of viruses by antibody
ii. Proliferation of *M. tuberculosis* in respiratory system

Q.2. B. Discuss any two of the following: **(04)**

i. Mechanisms of surface antigen variation in Influenza Virus
ii. Antigenic shift in trypanosomal variant surface antigen

Q.2.C. Justify the following: **(02)**

i. Immunity against *Plasmodium* infection is rarely acquired
ii. People suffering from neutropenia are susceptible to fungal diseases

Section II

Q.3.A. Answer any two of the following: **(06)**

i. State the difference between humoral and cell mediated immune responses
ii. State the properties of B-cell epitopes
iii. Summarize the major components of cytosolic pathway of antigen processing and presentation
iv. Write an **overview** on the "regulation of complement system".

Q.3.B. Justify the following: **(04)**

i. The onset of secondary immune response is more rapid than primary immune response.
ii. For effective complement response, the proenzyme needs to be cleaved by proteolytic enzymes for it to become effective.

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Q.4. A. Answer any two of the following: **(06)**

i. Name the mediators of early responses in asthma and state the major consequences of these mediators.
ii. Give a schematic diagram for Type III hypersensitivity.
iii. Describe **any one** of the autoimmune disease affecting nervous system.

Q.4.B. Explain the following statements: **(04)**

i. Rh factor incompatibility can result in RBC lysis.
ii. Type I diabetes mellitus is an autoimmune disease.

Q.5.A. Answer any two of the following: **(06)**

i. Write short note on clinical uses of monoclonal antibodies.
ii. Differentiate between precipitation and agglutination reactions.
iii. Write a shortnote on hybridoma technology

Q.5.B. Differentiate between the following: **(04)**

i. Active and passive immunization
ii. Attenuated and Killed vaccines